



Introduction to Game-Based Learning

Objectives of the Chapter

- To trace the emergence and development of Game-Based Learning (GBL) from ancient civilisations to modern educational contexts, providing readers with a comprehensive understanding of the relevance and transformative potential of GBL in diverse cultural and technological settings.
- To explore the theoretical underpinnings of GBL by exploring various educational theories such as constructivism, behaviourism, cognitivism and sociocultural perspectives. This chapter aims to explain how these theories can support the design, implementation and evaluation of effective game-based learning experiences with a focus on encouraging active learning, metacognition, critical thinking and collaborative interaction.
- To focus on providing a balanced perspective by critically evaluating the benefits and challenges associated with integrating GBL into educational settings. This chapter addresses current issues and provides insights and recommendations to guide educators and practitioners in realising the full potential of GBL.

INTRODUCTION

This introductory chapter begins an exploration of Game-Based Learning (GBL), a dynamic pedagogical approach that uses games to improve engagement and learning outcomes. We begin by clarifying a comprehensive definition of play and games and establishing a fundamental understanding of their central role in education. On this basis, we explore the experiential and interactive nature of play, reflecting the insights of respected scholars such as (Piaget, 1952). It also traces the historical evolution of GBL from ancient civilisations to modern excellence (Göbel et al., 2010). We also conduct a thorough evaluation of the advantages and difficulties associated with incorporating games into the field of education. Despite the inclusion of important studies such as Clark et al. (2016), which emphasise the potential advantages of GBL such as increased engagement, improved problem-solving skills and the promotion of critical thinking, it is also crucial to consider factors such as screen time and the importance of effective instructional design. An important part of our discussion focuses on the theoretical underpinnings of GBL, including insights from classic authors (Piaget, 1952; Vygotsky, 1978). We emphasise the synergy of constructivist and experiential pedagogy and games that enable students to actively construct knowledge, engage in metacognitive practice (Riva, 2022), and enhance critical thinking skills. Additionally, our research extends to analogue games such as board and card games that involve tactile, multisensory learning experiences. These metaphors illustrate connections between theoretical concepts and practical applications, contribute to interpersonal dynamics and enrich the broader discourse on GBL. In conclusion, this chapter provides an overview of GBL and summarises its importance, challenges, theoretical underpinnings and potential. This multifaceted understanding will enable educators, policymakers, researchers, and stakeholders to intelligently navigate the GBL landscape and promote innovative educational practices in diverse contexts.

Historical Evolution of Game-Based Learning

This chronological study traces the evolutionary trajectory and prospects of GBL within the wider landscape of educational theory, practice, and innovation. By tracing historical roots, highlighting technological developments, and anticipating future directions, it highlights the essential

role of GBL in fostering engagement, improving learning outcomes and promoting lifelong learning in an evolving educational ecosystem. The use of games such as educational instruments, spans various dimensions, such as space, culture and civilisation, sustainable importance, adaptation, exploration and reflection across diverse contexts and fields (Egenfeldt-Nielsen, 2004; Egenfeldt-Nielsen et al., 2020). Game-based education is integrated with principles, and epidemiological and educational epidemiology to promote dialogue, and interesting and practical knowledge (Prensky, 2003). GBL uses the intrinsic motivational aspects of games to enable active engagement, skill development, collaborative learning and problem-solving between participants, improved learning outcomes, retention and transfer across a variety of educational settings.

Tracing historical roots, exploring contemporary trends and charting future perspectives, this review explores how GBL can increase engagement, improve learning outcomes and promote lifelong learning in an evolving landscape of educational technology, innovation and human-centred design. The use of games for educational purposes dates back to ancient civilisations. In ancient civilisations, traditional games were used as educational tools to teach important cultural skills, values and knowledge to younger generations. Ancient civilisations such as Egypt, Greece, Rome, China, India, Mesopotamia and Central America recognised the educational potential of games and integrated them into religious rituals, cultural rituals and historical documents to develop knowledge, customs and worldviews (Parlett, 1999). Games such as Senet, Mehen, Go, Mancala, Chess, Checkers, Backgammon, Pachisi, Lubo, Shogi, Mahjong, Chaturanga and Alquerque have served as platforms to promote strategic thinking, tactical planning and ethical behaviour among players (Bell, 1979; Gobet et al., 2004). Games have been an integral part of various cultures for thousands of years, from Egyptian Senet to Chinese Go. For the Maya, these games were more than just entertainment; they served as rituals to connect the underworld with the gods. Winning such games was considered a sign of God's favour while losing sometimes led to sacrifices. The concept of the "magic circle", which suggests that games create unique spaces through clear rules and shared experiences, offers insight for contemporary educators.

These early efforts also paved the way for future innovations in GBL, encouraging researchers, educators and game developers to explore, experiment and improve game-based pedagogies, methodologies and

technologies across a broad landscape of theory, practice and educational games. Recognising the potential of games to promote experiential learning, collaborative problem-solving and lifelong skill development, all stakeholders in the education sector continue to use GBL as an innovative approach to enhance learner engagement and improve learning outcomes and overall development in the education sector—considering twenty-first-century skills and beyond.

Educational pioneers such as Maria Montessori, Friedrich Froebel and Seymour Papert advocated the integration of play-based learning approaches and emphasised experiential learning, active participation and holistic student development (Papert, 1994). Their innovative efforts have led to the development of educational games that respond to different learning styles, aptitudes and abilities that increase teaching effectiveness, student motivation and academic achievement in traditional educational settings.

A major development during this time was the development of board games designed to strengthen students' academic concepts, critical thinking and problem-solving skills (Klopfer et al., 2009). Games such as “Math Bingo” and “Scrabble” are examples of early attempts to stimulate interest, motivation and active participation in learning activities by integrating game mechanics, rules and objectives with educational content. Card games have also emerged as another popular form of educational play in formal educational settings that emphasise memory retention, pattern recognition and strategic planning (Prensky, 2003). Card games such as “Flashcard Matching” and “Vocabulary Go Fish” have helped students acquire and apply subject-related knowledge, and develop vocabulary and language skills, thereby improving academic achievement and skill acquisition.

Advances in computer technology in the late twentieth century facilitated the development of simple computer activities and educational software designed to reinforce students' academic concepts, cognitive skills and digital literacy (Gee & Gee, 2007). Educational software such as Math Blaster and Reader Rabbit are early attempts to use interactive multimedia, animated graphics and adaptive learning algorithms to create engaging, personalised and effective learning experiences in digital learning environments. The advent of digital technologies, including computers, game consoles and mobile devices, has revolutionised the landscape of GBL, providing interactive, immersive and adaptive learning

experiences that respond to different learning styles, preferences and abilities (Gee, 2003). Digital games such as Math Blaster, Oregon Trail and Civilization integrate educational content into game mechanics to foster creativity, collaboration and critical thinking among players.

The latest GBL trends also include Augmented Reality (AR), Virtual Reality (VR) and Mixed Reality (MR) technologies, enabling educators, designers and developers to create immersive, interactive and personalised learning that goes beyond traditional educational approaches, create an experience (Mcdowell et al., 2006). Combining game design principles, pedagogical strategies and technological innovation, GBL continues to evolve, adapt and innovate at the dynamic intersection of education, technology and human-centred design. The emergence of digital technologies, interactive platforms and virtual reality has revolutionised game-based learning environments and expanded opportunities and changes in formal, experiential and inclusive interconnected learning ecosystems (Egenfeldt-Nielsen et al., 2020; Gee, 2003; Gobet et al., 2004; Prensky, 2003; Sousa & Costa, 2018). Also in the twenty-first century, there is a paradigm shift in GBL based on the development of digital technologies such as computers, game consoles, mobile devices, AR, VR and MR platforms. These technological innovations have helped create engaging, interactive and adaptive learning experiences that go beyond traditional educational approaches and encourage creativity, collaboration, critical thinking and problem-solving skills among participants. The integration of simulation, VR and experiential learning environments in GBL also expands the scope and potential of game-based learning to provide realistic, contextual and personalised learning experiences that respond to a variety of situations, educational needs, goals and objectives (Mcdowell et al., 2006). Historically, games have been likened to important activities such as war, but the growing number of educational games on a variety of subjects have failed to demonstrate their effectiveness (Maratou et al., 2023). Gamification techniques, such as grading answers, have been incorporated into training to increase engagement, but this does not always improve content understanding or develop critical thinking skills. As a result, games have evolved into purposefully designed game-based learning interventions that are closely related to specific learning objectives, such as the use of analogue games for disaster management and decision-making (Rye & Aktas, 2022).

Empirical research has contributed to the dissemination of evidence-based practices in GBL (Gee, 2003; Göbel et al., 2010). The historical

background of GBL highlights the value and continued use of games as powerful educational tools that evolve over time and transcend cultural, technological and disciplinary boundaries to meet diverse learning needs, contexts and environments.

The game of ancient Egyptian councils, such as the Centre, promotes strategic thinking and decision technology. For example, we solve problems between traditional Wei-Chei critical thinking, space thinking and participants.

In the twentieth century, educational games, board games and simulations emerged as official tools to bolster academic concepts, enhance cognitive skills and develop social capabilities. Educational pioneers such as Maria Montessori and Friedrich Froebel emphasised the importance of students' experiential learning, active participation and balanced development and advocated the integration of play-based learning approaches. The beginning of GBL in the twentieth century saw the development of casual educational games, including board games, card games and simple computer activities designed to reinforce academic concepts and cognitive skills in formal educational settings (Clark et al., 2016). These early efforts laid the foundation for integrating game design principles, interactive dynamics and pedagogical strategies into educational environments to encourage student engagement, motivation and active learning. The twentieth century was a pivotal era for GBL, with the emergence of simple educational games that attempted to combine entertainment and education. This groundbreaking effort focuses on harnessing the unique appeal of games to promote the acquisition of academic concepts, cognitive skills and social and emotional competencies in a variety of educational contexts. The development and implementation of casual educational games in the twentieth century laid the foundation for incorporating game design principles, interaction mechanisms and pedagogical strategies into formal educational settings. By including elements such as rules, goals, objectives, feedback mechanisms and reward systems, teachers and instructional designers have sought to encourage student engagement, motivation and active learning, thereby improving student achievement, retention and learning outcomes.

WHAT IS A GAME?

A game is more than leisure. It involves a wide range of structured experiences designed to engage, educate and engage participants. At the heart of the game concept are elements such as rules, tasks and objectives that help shape the game experience. Games go beyond simple leisure activities and become multifaceted experiences that serve a variety of purposes, from entertainment to learning, participation and social interaction (Salen & Zimmerman, 2004). This comprehensive study explores the complexities of games and describes their defining characteristics, components and contributions to promoting creativity, collaboration and cognitive development among participants. Key to understanding games are the fundamental elements such as rules, challenges and goals that shape the game experience and contribute to the interaction, competition and cooperation among participants.

The core of a game consists of rules that serve as a structural framework for interaction, establish boundaries and promote meaningful interaction among participants (Caillois, 2001). These rules establish acceptable behaviour, interactions and outcomes and ensure fairness, clarity and consistency in the gaming environment. Rules provide a structured framework of play that allows players to solve problems, make informed decisions and interact with the game system and other participants in a consistent, coherent and cooperative manner. Games also include challenges, obstacles and tasks designed to stimulate cognitive, emotional and behavioural activity and develop participation, problem-solving and strategic thinking skills (Malone, 1981).

These challenges range from puzzles, quests and quests to races, battles and collaborations, allowing players to explore, experiment and achieve within the game environment. Offering varying levels of challenge, complexity and novelty, challenges maintain interest, encourage exploration, develop skills, enrich the overall game experience and encourage personal growth, resilience and adaptation for participants.

In addition, games include specific goals, objectives or outcomes that provide players with direction, purpose and motivation, guide progress, promote participation and promote satisfaction within the game context. These goals may include reaching certain milestones, mastering certain skills, achieving predetermined outcomes or aligning players' actions, decisions and strategies with desired results, rewards and outcomes.

Setting specific, actionable goals and challenges improves player motivation, focus and engagement, enriches the game experience, and fosters a sense of accomplishment, progress and satisfaction among participants. In conclusion, games are more than leisure. It includes a wide range of structured practices designed to engage, educate and involve participants in situations, domains and settings. By combining rules, tasks and responsibilities, games are a coherent, engaging and meaningful framework for stimulating creativity, collaboration and cognitive development among participants, enriching personal experiences, enhancing social interaction and promoting lifelong learning, growth and well-being.

Game vs Play in Educational Settings

Games and play serve a variety of functions in education by promoting active, experiential learning that promotes participation, motivation, participation, inclusion and global citizenship (Dewey, 1977; Freire et al., 2016; Kolb, 1986; Piaget, 1952; Vygotsky, 1978). Games are structured, rule-based activities, while play is a spontaneous, motivated activity. Below are definitions to differentiate between the two.

A game can be defined as structured, interactive, rule-based, goal-oriented and competitive, allowing players to engage in problem-solving, decision-making, communication and learning in a game with predefined rules, roles, tasks and constraints. This includes an activity or scenario that is realistic, engaging and engaging through mechanics and rewards, narratives, dynamics and environments (Gee, 2003; Rieber, 2005; Salen & Zimmerman, 2004). According to the TEGA project (Maratou et al., 2023), games are spontaneous, enjoyable activities, decoupled from the real world and structured with defined rules and outcomes or other quantitative feedback that allows players' performance to be reliably compared. Play is a free and meaningful activity for its own sake, separated in space and time from the demands of real life and bound by an autonomous system of rules that fulfils absolutes.

On the other hand, play is a spontaneous, intrinsically motivated and spontaneous activity that allows students to explore, experiment, take risks, develop skills, build relationships, build community and express themselves through experiential, participatory and creative activities, achieving goals and realising opportunities in a safe and supportive environment (Csikszentmihalyi, 1990; Huizinga, 1938/1949; Sutton-Smith, 1997).

Shaping the Design, Execution and Impact

As mentioned above, games are structured activities governed by a predetermined set of rules, goals and challenges designed to promote participation, learning and enjoyment by participants (Salen & Zimmerman, 2004). This structured system separates play from spontaneous or unstructured play and provides a clear framework within which players can interact, compete and cooperate. Essential to understanding games is the basic concept of “play”, which is characterised by voluntary participation, enjoyment and intrinsic motivation (Sutton-Smith, 1997). Play serves as a catalyst for creativity, exploration and personal growth in a structured play environment that allows participants to solve problems, make choices and make meaningful connections. To conceptualise and play a game in useful terms, it is essential to isolate the important components that shape its design, execution and impact.

Rules and Structure

Games are structured activities governed by a set of rules that establish boundaries, promote competition and cooperation and ensure fairness among participants. These rules serve as actionable guidelines that provide clarity, consistency and a common language for meaningful gaming experiences. A game is an internally structured entity that requires a specific set of rules to guide the actions, interactions and actions of participants within a defined setting. The main role of rules in a game can be divided into several important dimensions, each of which contributes to the overall effectiveness and experience of the game (Caillois, 2001). Rules serve as boundaries around which players operate and describe acceptable actions, interactions and strategies. These boundaries create order and predictability by creating a structured environment conducive to play and ensuring that participants understand their roles, responsibilities and limitations. Rules play an important role in shaping the dynamics of competition and cooperation between participants. By defining objectives, tasks and evaluation mechanisms, the rules create a competitive environment that encourages strategic participation. At the same time, rules can encourage teamwork, communication and cooperation by creating mutual support, shared ethics and common goals (Deterding et al., 2011). An important function of rules is to ensure fairness and equality among participants and to minimise potential bias or disagreement. The

rules establish consistent standards of play, preventing unfair advantages and maintaining the integrity of the game. In addition, rules maintain the legitimacy of the game by providing a basis for resolving disputes, clarifying ambiguities and reducing conflict. Rules serve as useful guidelines and provide clarity, consistency and structure to the game experience (Cailliois, 2001). According to contemporary perspectives—e.g. Salen and Zimmerman (2004)—these game rules provide a clear framework, guiding acceptable behaviours, interactions and strategies within specific game contexts. Importantly, rules uphold fairness by eliminating biases, resolving disputes and preserving the game’s integrity. They stimulate engagement by offering clear goals, incentives and outcomes, balancing player frustration with satisfaction and enhancing overall gaming experiences (Rigby & Ryan, 2011). Ultimately, a nuanced understanding and strategic application of these rule dimensions enable designers, educators and stakeholders to create compelling, immersive and resonant gaming experiences aligned with participants’ motivations and aspirations.

Goals and Objectives

In addition to rules, games include specific goals that provide direction, purpose and motivation to players (Malone, 1981). These actionable goals encourage participation, strategic thinking and problem-solving skills while encouraging satisfaction and achievement within the context of the game. In games, goals and objectives are important components that shape players’ participation, strategic awareness and overall experience in a given game environment. By defining tangible and intangible rewards, achievements or progress, these actionable goals stimulate player interest, ambition and commitment and encourage the pursuit of mastery, progress and achievement within the context of the game. These motivational dynamics reinforce player investment, persistence and satisfaction, enrich the overall gaming experience and contribute to lasting player loyalty. These objectives not only encourage strategic thinking and problem-solving but also foster satisfaction and achievement, driving players to interact more deeply with the game mechanics (Rigby & Ryan, 2011) and influencing their engagement and experience within the game environment (Deterding et al., 2011). By setting clear milestones, challenges and rewards, goals guide players’ actions and decisions, promoting coherent gameplay and intensive engagement (Huotari & Hamari, 2017). Moreover, these goals act as catalysts for intrinsic motivation, encouraging

players to pursue mastery, progress and achievement within the game context (Ryan & Deci, 2000) to reinforce player commitment, satisfaction and loyalty. Importantly, these objectives stimulate strategic thinking, adaptability and skill development, requiring players to employ analytical judgement, tactical planning and creative problem-solving (Csikszentmihalyi, 1990). By recognising achievements and progress, goals bolster players' self-esteem, competence and self-efficacy, further enhancing their satisfaction and motivation within the game ecosystem.

PLAYER PARTICIPATION

Games are meant to encourage active participation, involvement and immersion in the game environment (Rouse, 2005). Active engagement involves creating compelling stories, dynamic scenarios and interactive mechanics that sustain interest, encourage exploration and foster social interaction among players. Player participation is a cornerstone of game design and evaluation, encompassing the complex dynamics of participation, immersion and interaction in the game ecosystem. Detailed research on player engagement has revealed its multiple dimensions, effects and implications in creating immersive, engaging and socially enriching gaming experiences. Creating a compelling story that resonates with your players' interests, motivations and ambitions is key to engaging players. By incorporating complex stories, characters and settings, game designers can create emotional resonance, interest and investment in players, fostering a deep sense of connection, empathy and immersion in the game world. The appeal of this narrative stimulates players' imagination, creativity and curiosity, encouraging exploration, discovery and continued investment in uncovering stories, challenges and opportunities. In addition, engaging players requires designing dynamic scenarios that adapt, evolve and respond. It describes the player's actions, decisions and strategies. By incorporating variability, unpredictability and complexity into game scenarios, designers can develop stimulating, challenging and rewarding experiences that sustain interest, motivation and engagement over time. This adaptive design approach encourages players to experiment, innovate and adapt their strategies, encouraging a dynamic interaction of exploration, problem-solving and mastery in a variety of game situations and challenges. Action-oriented participation also depends on implementing interaction mechanisms that allow for meaningful interaction, feedback and collaboration among players. By integrating responsive controls,

mechanics and systems, designers can create fluid, intuitive and engaging gameplay experiences that encourage exploration, experimentation and social interaction. This interactive design paradigm fosters a collaborative, competitive and collaborative environment that promotes communication, teamwork and shared experiences and increases player engagement, satisfaction and loyalty within the gaming community. In addition, player involvement emphasises the promotion of social interaction, community building and shared experiences among players. By designing multiplayer modes, cooperative challenges and community features, designers can create an active, inclusive and supportive game ecosystem that encourages communication, collaboration and connection among diverse players.

This paradigm of social engagement promotes positive social dynamics, relationships and community in the wider gaming environment, increasing player retention, loyalty and satisfaction. In conclusion, player engagement serves as a critical factor in shaping, maintaining and enhancing the overall player experience throughout the game. By creating engaging narratives, dynamic scenarios and interaction mechanisms and encouraging social interaction, designers, educators and stakeholders can create engaging, rewarding and socially enriching game experiences that resonate with diverse audiences, preferences and aspirations.

OUTCOME-ORIENTED DESIGN

A useful definition states that game mechanics, goals and tasks are aligned with specific learning outcomes or behavioural goals (Gee, 2003). By combining pedagogy strategies, feedback mechanisms and evaluation tools, designers can develop existing experiences that will help develop technology, education and changes. Designing results is an important paradigm in the game design empire, the strategic adjustment of the game epidemiology, the results of the target learning or the behavioural. This comprehensive approach contributes to the development and behavioural changes in the development of educational principles, feedback mechanisms and evaluation tools in various educational and interactive contexts. The results of the final design are clearly read for specific learning, specific reading, adjusting play mechanisms, specific learning, specific reading, clear reading or local education. By carefully designing game mechanics, dynamics and systems that align with intended learning objectives, designers can create engaging, engaging and meaningful learning

experiences that enhance cognitive, emotional and behavioural engagement. This alignment promotes seamless integration of content, context and interaction and encourages a synergistic relationship between play and educational objectives that improves retention, application and transfer of knowledge, skills and abilities. Furthermore, outcome-oriented design integrates pedagogical strategies, teaching methods and learning theories to optimise the effectiveness and impact of educational games. This educational integration fosters genuine research, critical thinking and reflective practices, enhancing and optimising learning outcomes. It enables the development of skills in formative assessment, feedback mechanisms and efficiency metrics. The basis of this evaluation allows students to verify, measure and optimise the results of learning, experiments and environments through the process of design, evaluation and improvement of processes.

The results of the development of behavioural changes, the development of technology, development of technology and objectives, support and experience of learning the situation. Education, applications, use and designers develop students and focus, and students develop technology, competence and instability, allowing people who are close to the real world, difficulties, opportunities and situations. The resemblance of this turnover rate offers students a comprehensive, integrated and useful approach, which learns, adapts and prepares in various ways, professional and social. The conclusion is the basic basis of experimental, interesting and changing experiences of crime, interactive average and digital learning. By aligning game mechanics, pedagogical strategies, feedback mechanisms and assessment tools to specific learning outcomes or behavioural goals, designers, educators and stakeholders can create useful and relevant solutions that empower students, improve outcomes and enrich the educational experience. Interventions can be developed.

ANALOGUE GAMES

The focus on analogue games in this book stems from the recognition of their importance and unique characteristics in the gaming experience. Analogue games, including board games, card games, tabletop role-playing games and other non-digital formats, provide tangible, social and cognitive interactions in the real world that often differ from digital games. With a focus on analogue games, we seek to explore the effects, design complexities and player dynamics specific to these types of games.

Additionally, in an age dominated by digital technology, understanding analogue games can provide valuable insight into core game mechanics, player engagement strategies and fundamental principles that transcend technical mediums. Through this exploration, we seek to illuminate the multifaceted dimensions of analogue games and encourage appreciation, understanding and critical discourse about their relevance and evolving role in modern society. In this context, analogue games refer to games that use physical components such as boards, cards, dice, cues or other types of materials, rather than in a digital or electronic format. These games often require players to interact directly with these physical elements to determine outcomes, make strategic decisions and engage in game dynamics. The term covers a wide range of activities, including board games, card games, tabletop role-playing games and other board games or traditional games.

Analogue games, within the broader context of game research and design (Montola et al., 2009; Salen & Zimmerman, 2004), explore fundamental principles of game design. In “The Playbook Phenomenon”, which examines the evolution and cultural significance of analogue game formats, board games are highlighted for their advantages over digital games. These advantages include lower cost, ease of implementation, greater customisation options and higher levels of player engagement. Board games, unlike digital games, encourage player participation. This is because the players themselves must “enable” various game components for the game to work. Successful application of the board game approach to GBL needs requires consistent recognition of each student’s learning needs through GBL, which allows teachers to respond to these needs (Sousa et al., 2022). Gathering ideas from the field of analogue game education enriches the conversation by highlighting the unique and valuable qualities of board games, card games and role-playing games. These games offer numerous benefits, including realistic, immersive, multi-sensory, interactive and innovative learning experiences. This approach promotes experiential learning environments, communities and networks (de Freitas, 2006; Riva, 2022). The use of various analogue games, such as puzzles, strategy games and simulation games, further enhances these outcomes.

Distinctive Contributions of Analogue Games to the Educational Landscape

Analogue games, including board games, card games, board games, role-playing games, puzzles, strategy games and simulation games, make a unique contribution to the educational environment, reinforcing and complementing the wider discourse on learning based on game. Below are some unique aspects of analogue games that highlight their importance in promoting meaningful learning experiences and outcomes:

- Multi-log;
- Analogue game dynamics;
- Political simulations;
- Strategic planning;
- Epidemiology scenarios;
- Tactical decision-making;
- Work simulations;
- Rule-based learning;
- Role-playing interactions;
- Opinion formation;
- Feedback mechanisms;
- Result analysis;
- Experiential learning.

These elements combine to create a comprehensive educational tool that enhances learning by engaging students in interactive and immersive experiences (de Freitas, 2006; Gee, 2003).

Material functioning in analogue games, including the use of materials such as sculptures, shelves, metres and accessories, plays a crucial role in the educational environment. These materials support understanding, synthesis, decision-making and development within various contexts. They help build associations, form networks, create ecosystems and apply critical thinking technologies. Analogue games facilitate social interaction through cooperation, communication and compassion. They enhance understanding, connection and active participation, thereby strengthening social structures and networks. This interaction promotes competence, opportunities and values, fostering a supportive social environment.

Cognitive development is another significant benefit of analogue games, as they encourage assembly, creativity, integration, synchronisation and the synthesis of creative ideas (Papert, n.d.). Emotionally, analogue games offer fun, pleasure and interest while promoting innovation and emotional regulation. They enhance emotional competencies and help players manage their emotions effectively (Rigby & Ryan, 2011).

By adopting useful definitions of games and play, educators, designers and stakeholders can create engaging and meaningful learning experiences (Prensky, 2003). Utilising these principles allows practitioners to design, evaluate and improve games to meet diverse learning needs, foster collaboration and achieve meaningful outcomes in various educational contexts.

We have previously evaluated the effectiveness and potential of analogue GBL in educational contexts (Sousa et al., 2022). The results show that analogue GBL serves as an effective teaching tool that affects learning, cognitive and psychological levels. Board games in particular are associated with high interest, satisfaction, flexibility and freedom of experience among students. It also helps develop soft skills such as learning, behaviour change, communication, collaboration, creativity, empathy and well-being. Furthermore, despite the inclusive nature of the board game community, research on analogue GBL has paid limited attention to issues of access and inclusion, and future research lacks a coherent framework linking populations, learning outcomes, game characteristics and larger social impact.

Advantages of Analogue Games

At the heart of all games is the basic principle of interactivity, which is characterised by the active participation, participation and interaction of participants in the game environment (Rouse, 2005). This interactive dynamic goes beyond mere viewing and encourages players to solve problems, make decisions and collaborate with others, encouraging immersion, creativity and social interaction. The inherent interactivity of games differentiates them from passive forms of entertainment or learning that emphasise dynamic participation, experiential learning and interactive communication between participants.

In addition, the interactive principles of games have a profound cognitive and behavioural impact that encourages critical thinking, problem-solving and decision-making skills in participants. By encouraging

active participation, exploration and experimentation, games promote skill development, knowledge acquisition and behavioural adaptation, enhancing cognitive flexibility, creative problem-solving and strategic planning in a variety of situations and environments.

Games come in a variety of platforms, including board games, card games, role-playing games and digital games, each offering unique experiences, mechanisms and opportunities for participation (Koster, 2013). For example, traditional board games like Chess emphasise strategic planning, negotiation and resource management, while digital games like Minecraft promote creativity, exploration and cooperation in vast virtual worlds.

Additionally, each gaming platform has different dynamics, interactions and characteristics that shape gaming experiences, player roles and opportunities for collaboration in different situations and environments. Using platform-specific mechanics, events and challenges, designers can develop immersive, engaging and meaningful experiences that resonate with players' interests, motivations and aspirations, fostering creativity, collaboration and social interaction across platforms and environments.

In addition to entertainment and activity, games have an important psychological and social impact that shapes participants' cognitive development, social skills, emotional regulation and overall well-being (Granic et al., 2014). The structured yet flexible nature of games considers individual preferences, abilities and learning styles, enabling personalised experiences that improve engagement, satisfaction and performance in a variety of personal, educational and professional contexts.

Games also help achieve meaningful outcomes by fostering resilience, adaptation, empathy and cooperation among participants, thereby encouraging positive social interaction, mutual respect and shared understanding in diverse communities and societies. Combining psychological principles, social dynamics and educational strategies, games are a comprehensive way to develop important life skills, promote well-being and foster an inclusive, supportive and collaborative environment in an evolving interactive media environment, digital technologies and people.

Another study found that analogue GBL provides significant benefits and develops important soft skills such as collaboration, communication, creativity, problem-solving, and decision-making (Maratou et al., 2023; Sousa et al., 2023b). It also provides flexibility, adapts to specific classroom needs, strengthens teacher-student relationships, and develops individual skills. However, two major problems arise: this means that

the games take a long time to create and must be customised for the individual needs of the class. This can lead to significant financial and time costs. Furthermore, this study highlights the limited use of quantitative assessment methods in GBL, with an emphasis on observational assessments due to time constraints. In terms of inclusion, GBL offers the potential to encourage collaboration and break down barriers, but there are accessibility concerns and cultural considerations for students with learning disabilities. This study is to carry out institutional support, education, and cooperation between experts and the academic environment, competitions, experts and experts to improve the large and efficient efficiency of analogue GBL in the context of higher education and comprehensive measures in the GBL.

Some authors—as Granic et al. (2014) or Koster (2013)—synthesise the concept of interaction, playground and psychological and social consequences and interactive experiments in the media, social interaction of social interaction emphasises the integrated role. In the modern environment of social interaction, educational environment, digital technology and design, the basis of each game is the principle of interactions in which participants actively enter the game environment, epidemiology and other players (Rouse, 2005). This is a special game of interactive dynamic stimuli, cooperation and social interactions, entertainment or passive entertainment or training.

BENEFITS AND CHALLENGES OF USING GAMES IN THE LEARNING ENVIRONMENT

GBL refers to a teaching method that includes game elements to facilitate the learning experience, making the educational process more interesting, interactive and engaging for students. This pedagogical strategy uses game principles such as rules, tests, rewards, feedback mechanisms and goal-oriented tasks to develop students' active participation, problem-solving skills, critical thinking, cooperation and motivation (Gee, 2003; Prensky, 2003). By incorporating game elements into educational contexts, educators can create dynamic learning environments that promote deeper understanding, retention and application of content knowledge while responding to different learning styles, preferences and interests (Egenfeldt-Nielsen et al., 2020; Egenfeldt-Nielsen, 2004, 2006). GBL includes a variety of formats, including digital games, board games, simulations and role-playing, tailored to specific learning objectives,

curriculum standards and pedagogical goals. Through strategic design and implementation, it provides innovative opportunities to increase engagement, motivation and achievement in educational settings with an emphasis on experiential learning, exploration and discovery (Dickey, 2011; Squire & Jenkins, 2011).

GBL improves activity, problem-solving and critical thinking. Analogue games provide hands-on experiences that connect theoretical concepts to real-world scenarios, solving complex ideas and promoting learning. These games encourage critical thinking, problem-solving and analytical skills while actively engaging students in meaningful ways. It is consistent with social constructivist learning theory, which promotes understanding by placing information in a recognisable context and emphasises social interaction and cooperation. In addition, the experience of supporting analogue games is to use a multidimensional approach to train reflections and meta spokes for better understanding. They offer dynamic approaches to the team to develop and learn the technology of the autonomous government through active participation and various experiences (Rye, & Sousa, 2023). GBL enhances the student experience by engaging, motivating and empowering students by promoting active participation, participation and change in the educational environment (Egenfeldt-Nielsen, 2006; Gee, 2003; Prensky, 2003). It also develops students' critical thinking, logical thinking, analytical skills and application skills, giving students the skills and competencies they need to solve complex problems and succeed in a variety of situations. Moreover, it also promotes cognitive development, metacognitive awareness, intellectual growth and orientation that promotes students' critical thinking and reflective practice. In modern educational paradigms, the concepts of experiential and interactive learning are cornerstones that emphasise the importance of direct experience, participation and relationships in developing meaningful learning experiences (Dewey, 1977; Illeris, 2009; Kolb, 1986). In this context, games emerge as powerful pedagogical tools to engage, motivate and empower students, allowing them to actively engage with content and experiences in dynamic and relevant ways. This requires effective instructional design, which requires investment in support, training, professional development and capacity building among designers, developers and educators (Clark et al., 2016; Mayer et al., 2014; Squire & Jenkins, 2011). To explore the elements and benefits of GBL, it is useful to first think about what the game is.

Benefits of Game-Based Learning

GBL offers several benefits that enhance the educational experience and promote effective learning outcomes. First, GBL promotes engagement and motivation by using interactive and engaging game elements to engage students and keep them focused during the learning process. This active participation often leads to greater engagement, active participation and intrinsic motivation to achieve learning goals. Second, GBL promotes experiential learning by helping students develop critical thinking, problem-solving and decision-making skills by providing opportunities to explore, experience and apply knowledge in simulated or real worlds. GBL also promotes personalised learning experiences tailored to individual learning styles, preferences and pace, meeting the diverse needs of learners and promoting a supportive and inclusive learning environment. In addition, GBL encourages collaboration, communication and teamwork among students, encouraging the development of social interaction, collaboration and interpersonal skills essential for academic and professional success. Overall, GBL improves learning effectiveness, engagement, knowledge retention and application through interactive, immersive and experiential learning experiences. Let's take a look at some of the key benefits below.

Engagement and Motivation

Games are primarily designed to motivate and engage students through a combination of immediate feedback, adaptive challenges and personal experiences tailored to individual skill levels (Gee, 2003). This design philosophy resonates with the tenets of self-determination theory, which states that autonomy, competence and relatedness are crucial factors in promoting intrinsic motivation (Ryan & Deci, 2000). The game's immediate feedback mechanism allows students to measure their performance, identify areas for improvement and adjust their strategies, accordingly, increasing their confidence and self-esteem (de la Barrera et al., 2021). In addition, adaptation to the game is always convinced that students have a regular complexity, so they are balanced to maintain the "flow" situation, which is the concept of current coordination and implementation.

Personal publications that meet educational games, meetings and requirements are intended for education, so we meet the diverse needs and education requirements (Plass et al., 2015). This is an individual Universal Design for Learning (UDL) approach that emphasises the

needs and preferences of the students. Social cognitive theory, energy theory and universal design, along with active cooperation, permanent participation and active practices, form the foundation of GBL-Comers' professional development platform. This platform is crucial in the twenty-first century, fostering critical thinking, problem-solving, cooperation, communication, creativity and digital literacy.

In the context of developing competencies, it's essential to cultivate the capacity for integrating games and technology into learning. This involves enhancing students' abilities and skills through challenging, game-based and interactive tasks. This method aligns with Kolb's experiential learning theory (Kolb, 1986), which describes learning as a cyclical process that includes concrete experience, reflective observation, abstract conceptualisation and active experimentation. In addition, the collaborative nature of many educational games promotes social interaction, teamwork and interpersonal skills, consistent with Vygotsky's theory of social development (Vygotsky, 1978) which emphasises the role of social interaction and cultural context in cognitive development and learning.

In summary, GBL offers a multifaceted approach to skill development that aligns with contemporary frameworks such as collaborative and experiential learning theory and theoretical perspectives such as social development theory for twenty-first-century skills. Encouraging critical thinking, problem-solving, collaboration, communication, creativity and digital literacy, GBL provides students with the critical competencies and skills they need to navigate and succeed in today's complex, interconnected and rapidly changing world.

Personalised Learning

The adaptive learning mechanisms inherent in game-based environments promote personalised learning tailored to an individual's learning preferences, style, needs and pace (Squire, 2006). Using advanced algorithms, analytics and artificial intelligence, educational games dynamically adjust content delivery, difficulty and learning strategies to enhance the learning experience, promote retention and promote learning for diverse student populations.

In the context of analogue games, adaptive learning can be achieved through flexible game design and inclusive practices. By incorporating advanced design principles and adaptive strategies, analogue games can dynamically adjust the complexity and engagement level to suit different players. For example, analogue games can include varied difficulty levels,

customisable rules and modular components that can be adjusted based on player feedback and preferences. For instance, board games can have adjustable scenarios or missions that change based on the players' performance and decisions, similar to how a digital game might alter its narrative or challenge based on real-time analytics. Card games can introduce new rules or challenges as players' progress, ensuring that the game remains engaging and educational for diverse student populations. Therefore, the principles of adaptive learning can enhance the educational value of analogue games by promoting retention, engagement and learning among players with different needs and preferences. By leveraging innovative design approaches and inclusive practices, analogue games can offer personalised and effective learning experiences similar to their digital counterparts. This adaptation is consistent with the principles of differentiated instruction, a pedagogical approach that emphasises personalising content, processes and outcomes based on students' unique learning profiles (Tomlinson, 2001). Using advanced algorithms, analytics and artificial intelligence, educational games dynamically adjust content delivery, difficulty and learning strategies (Baker & Inventado, 2018).

This adaptation is consistent with Vygotsky's (1978) concept of the zone of proximal development (ZPD), allowing for real-time feedback and corrections that ensure students are challenged but not overwhelmed. In addition, personalised learning in the game-based environment is a key factor in internal stimuli and participation, which corresponds to autonomy, independence, competence, relationship and self-determination (Ryan & Deci, 2000). The selection, support and situation of students and adaptive educational games help to increase the feeling of autonomy, competence, encouragement, interest and learning (Ifenthaler & Yau, 2020).

In conclusion, adaptive learning mechanisms embedded in game-based environments, supported by advanced algorithms, analytics and artificial intelligence, provide a personalised, engaging and effective learning experience. These adaptations are consistent with established educational theories and frameworks, including differentiated instruction, ZPD, self-determination theory and data-driven decision-making approaches, to improve learning experiences and outcomes for diverse students. With analogue games, having a game facilitator with design skills allows for real-time, low-cost adaptation (Sousa et al., 2021).

Therefore, GBL offers many advantages, but there are some disadvantages that teachers and students should be aware of. First, implementing

GBL requires significant resources, including time, money and technical infrastructure, which can be challenging for organisations with limited resources or expertise. Second, designing effective educational games that meet specific learning objectives, curriculum standards and learning objectives is complex and time-consuming, requiring collaboration among teachers, instructional designers and game developers. In addition, GBL can be distracting as students tend to focus more on game elements than learning outcomes, which can undermine the educational effectiveness of the learning experience. Additionally, GBL may not meet the needs and preferences of all learners, as some individuals may not be motivated or engaged in game-based approaches and may prefer traditional learning methods. Additionally, assessing student progress, performance and content mastery can be difficult, and innovative assessment strategies and tools are needed to accurately measure learning outcomes.

Challenges of Game-Based Learning

Although GBL provides an innovative and engaging learning experience, educators must consider its limitations and challenges to effectively integrate play-based approaches into general education settings.

Barriers to Implementation

Integrating GBL into general education requires addressing many logistical, technical, administrative, pedagogical and institutional challenges (Egenfeldt-Nielsen, 2006).

These barriers include infrastructural constraints, limited resources, curriculum coordination, teacher training, policy formulation and stakeholder resistance that complicate the adoption, implementation and sustainability of play-based pedagogies in formal education contexts (Sousa et al., 2023a). The adoption and integration of GBL in traditional educational settings pose multifaceted challenges that include logistical, technical, administrative, pedagogical and institutional dimensions. Traditional institutions may lack the necessary technological infrastructure, including compatible devices, reliable Internet connections and appropriate software platforms, needed to support immersive and interactive GBL experiences (Johnson et al., 2016).

Developing and implementing high-quality educational games requires significant investment in terms of time, money, expertise and professional resources, including game developers, instructional designers, content

experts and evaluators (Kebritchi et al., 2010). The alignment of GBL activities with established curriculum standards, learning objectives and assessment criteria poses challenges for the consistency, relevance, reach and progress of the educational programme. Teachers need professional training, professional development and ongoing support to effectively integrate play-based pedagogy into their teaching practices, assess learning outcomes and promote meaningful learning experiences for students (Plass et al., 2015).

Implementation of a GBL initiative requires the development and adoption of supportive policies, guidelines and frameworks at the institutional, regional, state or national level to address regulatory, ethical, legal and compliance issues (Becker, 2011). Administrators, teachers, parents, politicians and other stakeholders may arise due to concerns about the effectiveness, equity, accessibility, acceptability, sustainability and cultural appropriateness of the GBL approach (Sousa et al., 2023a). In short, implementing GBL in the context of formal education requires navigating a complex environment characterised by infrastructure constraints, limited resources, curriculum coordination issues, teacher training needs, policy considerations and stakeholder resistance. Addressing these challenges requires a collaborative, systematic and strategic approach that harnesses the potential of play-based pedagogy by integrating technology investments, pedagogical innovations, policy reforms, professional development initiatives, stakeholder engagement strategies and evidence-based practices.

Assessment and Evaluation

Assessing and evaluating learning outcomes in game-based environments presents unique complexities that require the development and adoption of innovative, valid, reliable and context-appropriate assessment strategies tailored to game dynamics, objectives and teaching methods (Wiley, 2002). Traditional assessment methods do not adequately reflect the multifaceted, dynamic and nuanced nature of learning experiences in game-based contexts, so alternative assessment approaches, tools and frameworks need to be explored and implemented. Assessing and evaluating learning outcomes in game-based environments introduces unique challenges and complexities beyond traditional assessment paradigms including complex learning dynamics, comprehensive learning outcomes, contextual relevance, alternative assessment approaches, validity and reliability.

GBL environments are inherently dynamic, interactive and adaptive, providing students with a variety of pathways, options, tasks and experiences that may not fit within standardised assessment measures (Gee, 2003). Therefore, assessing student proficiency, performance and progress requires a comprehensive understanding of game mechanics, interactions and feedback mechanisms. Games help develop and acquire a wide range of cognitive, emotional, social and behavioural skills, including problem-solving, critical thinking, collaboration, creativity, decision-making, emotional regulation, persistence and adaptability. Traditional assessment methods such as standardised tests, quizzes and exams may not capture the richness, complexity and multifaceted nature of these learning outcomes. Assessment of learning in game-based environments requires contextually relevant, authentic and meaningful assessment strategies that are consistent with specific objectives, pedagogical approaches, learning models and curriculum (Ifenthaler & Yau, 2020).

Considering the limitations of traditional assessment methods, educators, researchers and designers explore and implement alternative assessment approaches, including formative assessment, performance-based assessment, portfolio assessment, peer assessment, self-assessment, analysis of the game and evaluation, modelling, scenarios, stories and empirical evaluations (Scouller, 1998). Ensuring the validity, reliability, fairness, fairness, accessibility and ethical integrity of evaluation practices remains a major concern. Developing valid and reliable tools, instruments, rubrics, criteria, benchmarks and assessment protocols that align with game dynamics, goals, pedagogies and standards requires a rigorous process of research, development, validation and refinement (Black & Wiliam, 2010; Shute & Rahimi, 2017).

In summary, assessing learning outcomes in game-based environments poses unique challenges related to complex learning dynamics, multifaceted outcomes, contextual relevance, alternative approaches, validity and reliability. Addressing these challenges requires a collaborative, interdisciplinary, evidence-based approach that integrates pedagogical insight, technological innovation, assessment expertise, stakeholder engagement and continuous improvement efforts to improve the effectiveness, efficiency, equity and reliability of assessment practices within GBL ecosystems.

Sustainability

Designing, developing, implementing and maintaining high-quality educational games require significant investment in terms of time, resources, expertise, research, development and ongoing support (Kebritchi et al., 2010). Concerns about long-term sustainability, scalability, cost-effectiveness, accessibility, relevance and return on investment can discourage educators, institutions, developers and policymakers from adopting and investing in GBL initiatives, limiting its potential impact, scope and effectiveness (Sousa et al., 2023a).

The sustainability of GBL initiatives remains a critical issue for educators, institutions, developers, policymakers and stakeholders. Several factors contribute to the challenges and considerations associated with ensuring the long-term viability, effectiveness, scalability, accessibility and impact of educational games in diverse learning environments. Developing and distributing quality educational games requires significant investment in financial resources, human capital, technical infrastructure, instructional design expertise, content development, user experience design, testing, evaluation, iteration and ongoing maintenance (Fabricatore & López, 2014; Maratou et al., 2023; Sousa et al., 2023a). These resource-intensive efforts require sustainable funding models, grant opportunities, partnerships, collaborations and innovative funding mechanisms to support long-term development and expansion efforts.

Integrating GBL initiatives into formal education curricula, instructional practices, learning environments, assessment strategies and professional development programmes requires alignment, collaboration, coordination, communication and commitment among educators, administrators, policymakers, developers, researchers and stakeholders. To sustain learner engagement, motivation, participation, satisfaction, retention, progression and success within GBL environments, it is essential to design, implement, evaluate, refine and enhance immersive, interactive, adaptive, challenging, rewarding and meaningful experiences. These experiences should resonate with the diverse preferences, needs, interests, abilities, backgrounds and contexts of learners (Dickey, 2011).

Advocating for evidence-based practices, research-driven approaches, empirical studies, rigorous evaluations, comparative analyses, longitudinal studies, meta-analyses and best practices within the field of GBL is essential to inform decision-making, inform policy development, guide instructional design, shape pedagogical strategies and foster continuous improvement efforts (Wouters et al., 2013). In summary, ensuring the

sustainability of GBL initiatives requires a multifaceted, collaborative, interdisciplinary, evidence-based, strategic, adaptive and iterative approach that addresses the complexities, challenges, opportunities and considerations associated with resource allocation, technological innovation, pedagogical integration, user engagement and empirical research within broader educational ecosystems.

LEARNING THEORY IN THE CONTEXT OF GAME-BASED LEARNING

Learning theories research can advance our understanding of GBL by providing theoretical frameworks and insights into how people acquire, process and apply knowledge in educational contexts. By examining key theories such as behaviourism, cognitivism, constructivism and sociocultural perspectives, educators and researchers can identify key principles, mechanisms and processes that influence student engagement, motivation, cognition and behaviour in play environments. For example, behaviourist theory emphasises the role of reinforcement, feedback and repetition in shaping student behaviour and skills, which can support the design of effective game mechanics, reward systems and feedback mechanisms in educational games. Cognitive theory also seeks to understand students' mental processes, memory, problem-solving strategies and information-processing skills and provides insight into designing games that promote cognitive development, critical thinking, problem-solving and metacognitive awareness. In addition, constructivism and sociocultural theory emphasises the importance of active, collaborative, experiential and context-based learning experiences, and GBL allows students to explore, experiment, collaborate and develop knowledge in meaningful, authentic and interactive learning environments.

By integrating insights from a variety of learning theories, educators and designers can develop more engaging, effective and pedagogically sound GBL experiences that meet students' needs, preferences and learning goals—as suggested in Table 2.1. GBL learning theory emphasises the relationship between games and constructivist and experiential pedagogy to promote meaningful, stimulating and holistic learning experiences (Egenfeldt-Nielsen, 2006; Gee, 2003; Papert, 1994; Piaget, 1952; Vygotsky, 1978).

Table 2.1 Theoretical foundations of GBL design

<i>Aspects/ theories</i>	<i>Behaviourism</i>	<i>Cognitivism</i>	<i>Constructivism</i>	<i>Sociocultural theory</i>
Contribution to play-based learning	Emphasises reinforcement, feedback and repetition in shaping behaviour and skills	Focuses on mental processes, memory, problem-solving strategies and information-processing skills	Highlights active, collaborative, experiential and context-based learning experiences	Emphasises learning through social interaction, cooperation, communication and community development
Key principles and mechanisms	Reinforcement, feedback, repetition	Mental processes, memory, problem-solving	Active learning, collaboration, reflection	Social interaction, cooperation, modelling
Application in GBL design	Effective game mechanics, reward systems, feedback mechanisms	Games that promote cognitive development, critical thinking, problem-solving	Designing activities that enable active participation, social interaction, reflection	Promoting collaborative learning, social interaction and community development
Influence on student engagement, motivation, cognition and behaviour	Shapes behaviour and skills through reinforcement and repetition	Enhances cognitive skills and problem-solving abilities	Encourages active participation, reflection and collaboration	Fosters social engagement, cooperation and communication

Source Author's elaboration

Building on works of Piaget (1952) and Vygotsky (1978), constructivist learning theory emphasises that students actively construct knowledge, meaning and transformative experiences in sociocultural, experiential, participatory and ecological environments. By integrating games into constructive learning environments, teachers can encourage, support and assess students' progress in their unique and evolving ZPD, thereby increasing cognitive skills and responsibility (Vygotsky, 1978; Wertsch, 1988).

Experiential learning theory, formulated by scholars such as Kolb (1986) and Dewey (1977), emphasises that students acquire knowledge, skills, competencies and scope. Integrating games into experiential learning environments promotes student reflection on personal, academic and lifelong learning situations and environments. These concepts are discussed in more detail below.

Active Knowledge Construction, Metacognition and Critical Thinking through Games

Games can serve as active learning tools that allow students to engage in interactive, experiential, experiential and collaborative learning environments in a variety of settings (Gee, 2003). Metacognition—defined as knowing and controlling one’s own cognitive processes—plays a significant role in improving students’ self-regulation and self-motivation (Flavell, 1979; Riva, 2022; Salen & Zimmerman, 2004). Games promote student engagement in reflective practice that can be analysed and evaluated in a collaborative and transformative learning environment (Moon, 2013; Riva, 2022; Schön, 2017). Games strengthen students’ critical thinking skills and function as a catalyst for the development of analytical habits, behaviours and evolving learning situations (Riva, 2022).

Piaget’s Perspective on Active Learning

Renowned developmental psychologist Jean Piaget (1952) formulated a comprehensive theory of cognitive development that emphasises the importance of experiential, interactive and constructive learning experiences in improving students’ intellectual growth, conceptual understanding, logical reasoning, problem-solving skills, metacognitive awareness and cognitive skills. Piaget (1952) emphasised the importance of active exploration, discovery and internalisation to promote cognitive-behavioural development through meaningful, stimulating and enriching environments. Students can explore the theoretical foundations of dynamic and constructive educational environments to develop skills such as creativity, critical thinking, emotional intelligence and lifelong learning (Prensky, 2003). By studying games through practical approaches and dialogue, we can assess students’ innovative potential in various educational contexts. Promoting internships and practical methods allows

us to evaluate the effectiveness of entertainment strategies in different educational situations.

Constructivist Learning Theory in GBL

How can one effectively incorporate knowledge, perception, understanding, significance and technological advancements into the theory of constructive learning as proposed by Piaget (1952) and Vygotsky (1978)? Investigation, understanding, technology, qualification, competence and expertise are essential elements in creating significant, relevant and realistic learning experiences. These involve social interaction, cooperation and collaboration with experts and associations. Through reflection, metacognition, self-regulation, self-evaluation, feedback, repetition, adaptation and modification, students can monitor, evaluate, adjust and develop their cognitive, affective, behavioural and metacognitive processes, strategies and tools. This optimisation results in a dynamic, complex and evolving learning environment (Costa et al., 2016; Schön, 2017; Sousa & Costa, 2018).

In GBL environments, students engage in problem-solving, research and collaboration. They immerse themselves in real, contextual and complex scenarios, tasks, problems, puzzles and projects that require critical thinking, decision making, creativity, innovation, resilience, persistence, adaptability and exploration. These activities take place in collaborative, competitive and cooperative contexts (Gee, 2003; Papert, 1994).

GBL environments encourage students to explore game mechanics, aesthetics, events, themes, rules, roles, challenges, goals, strategies, tactics, tools, techniques, interfaces, practices and outcomes. They promote collaboration, communication, negotiation, cooperation, coordination, teamwork, leadership, mentoring, coaching, guidance, support and sharing among students, peers, faculty, experts, stakeholders and communities to achieve common goals and outcomes (Dillenbourg, 1999; Steinkuehler & Duncan, 2008).

Constructivist learning theory supports the design and evaluation of activities that enable active participation, social interaction, reflection, metacognition, problem-solving, decision-making, inquiry, collaboration, communication and empowerment in a diverse, inclusive and sustainable educational environment.

Cognitive Load Theory and Its Application to GBL

Cognitive load theory emphasises the importance of managing and optimising cognitive resources, attention, working memory, schema activation and information processing during learning activities (Mayer & Moreno, 2003). Learning games are designed to match students' cognitive abilities, prior knowledge, experience level and learning goals to reduce external cognitive load, mitigate internal cognitive load and optimise relevant cognitive load to promote a learning experience efficient, effective and engaging (Hawlitshchek & Joeckel, 2017; Mayer & Moreno, 2003).

Cognitive Load Theory (CLT) provides a framework for understanding how limitations in human cognitive processing capacity affect learning, memory, transfer and application in educational contexts (Mayer & Moreno, 2003; Sweller, 1988). Working memory has a limited capacity and duration to process, store, retrieve and manipulate information, concepts, principles, procedures, strategies and practices during learning activities (Sweller, 1988). Cognitive load includes: internal load (related to the complexity of the learning material); external load (related to the design of the learning environment); and associated load (related to the organisation and integration of new knowledge into existing systems). Effective instructional design, pedagogical strategies, multimedia principles and learning environments can promote learning, retention, transfer and application by minimising extrinsic cognitive load, managing intrinsic cognitive load and maximising appropriate cognitive load.

Applications for the GBL environment primarily include adaptive difficulty, scaffolding and interactive design. These aspects involve setting appropriate difficulties, tasks, objectives, feedback, instructions and resources to simplify and enhance learning experiences. They support regulation and impact practice (Wouters et al., 2013).

Students benefit from managing cognitive loads, modelling, coaching, coordination, feedback and reflection (Wirth et al., 2020). Improving interactive principles, dynamics, mechanics, objectives, strategies, practices, technologies, interfaces and participation is crucial for creating meaningful, relevant and realistic learning situations (Gee, 2003; Moreno, 2010).

CLT offers valuable insights for the development, implementation and evaluation of concepts, principles, strategies and practices. It includes assessing and repeatedly evaluating these processes to optimise learning outcomes (Hawlitshchek & Joeckel, 2017; Mayer & Moreno, 2003).

Self-Determination Theory

Self-determination theory (SDT) emphasises the importance of autonomy, competence, relationships, intrinsic motivation, volition, self-regulation and satisfaction of psychological needs in promoting meaningful, motivating and transformative learning experiences (Ahmadi et al., 2023; Ryan & Deci, 2000; Ryan et al., 2021). GBL environments provide students with agency, choice, control, ownership, responsibility, challenge, feedback, support, collaboration, competition, recognition and rewards, which increase intrinsic motivation, participation, persistence, satisfaction, achievement and well-being (Przybylski et al., 2014; Ryan et al., 2006).

SDT provides a comprehensive framework for understanding how motivation, agency, autonomy, competence, relatedness, volition, self-regulation and psychological need satisfaction influence learning, development, growth and well-being in educational contexts. Students have intrinsic motivation, volition, agency, choice, control, ownership, responsibility and independence in achieving learning goals, assignments, tasks, challenges, activities, projects, assessments and reflections in a GBL environment. Learners develop domain-specific expertise, skills, knowledge, understanding, mastery, performance, mastery, performance and success in a GBL environment through engaging, interactive, immersive, challenging, rewarding and reflective experiences.

In the GBL environment, students build meaningful and authentic relationships, connections, interactions, communities, networks and cultures that are collaborative, supportive, respectful, inclusive, fair and empathetic. SDT provides valuable insights for designing, assessing and evaluating GBL environments that promote meaningful and stimulating learning experiences by supporting student autonomy, intrinsic motivation and engagement in an inclusive, equitable and accessible educational setting.

As previously discussed by Sousa et al. (2023b), analogue games, lacking automation systems, rely heavily on player agency and self-determination for activation, play and resolution of the game system.

Social Learning Theory

Social learning theory emphasises the role of learning through observation, modelling, imitation, social interaction, cooperation, communication, feedback, support and association in promoting cognitive, emotional and behavioural development in social contexts (Bandura, 2000; Vygotsky, 1978). GBL environments influence learning experiences, outcomes and environments by promoting social engagement, social activity, social interaction, social learning, social cohesion, social support, social influence and social networking among students (Dillenbourg, 1999; Gee, 2003).

Social learning theory in learning games provides a comprehensive framework that emphasises the importance of social interaction, collaboration, communication, supervision, modelling, feedback, support, community and context in shaping cognitive, emotional and behavioural outcomes in educational settings. The basic principles of social training theory are the following. Students in GBL environments are expected to engage in imitation, mentorship and instruction from peers, mentors, instructors and experts. They interact with characters, avatars and each other to develop behaviours, attitudes, perspectives, opinions and values. Through these activities, students are expected to create, sustain and promote inclusive, equitable, accessible, supportive, respectful, safe, empathetic and engaged learning communities, networks, cultures, environments and ecosystems. Their participation is crucial in driving change within the GBL environment (Nadolny & Halabi, 2016; Wenger, 2000).

CONCLUSION

In this chapter, we journey into the diverse world of GBL, starting with a fundamental understanding of games and play in educational settings. Building on the seminal work of scholars such as Piaget (1952) and Vygotsky (1978), we explore how games and play converge and diverge in their pedagogical implications, emphasising their role in fostering curiosity, creativity and learning experiences. Innovative education: it traces the historical development of GBL and demonstrates its relevance and adaptability in different contexts. From ancient civilisations to modern digital platforms, the innovative potential of GBL has emerged as a catalyst for experiential, immersive, and holistic learning experiences. Furthermore, our study highlighted the complex interplay between GBL

and mainstream learning theories (from behaviourism to sociocultural perspectives) and illuminated the mechanisms by which games improve cognition, cooperation and motivation. Central to this discourse was the recognition of GBL's ability to encourage active learning, metacognition and critical thinking, based on constructivist pedagogy and consideration of cognitive load. In exploring the landscape of GBL, insights from self-determination theory and social learning dynamics have further enriched our understanding, highlighting the importance of autonomy, communication and collaboration in shaping innovative educational practices. However, integrating GBL into the educational ecosystem is not without its challenges. Infrastructure limitations, curriculum coordination, teacher training and stakeholder resistance require a strategic and collaborative approach to effectively realise the potential of GBL. But despite these complexities, GBL's groundbreaking promise is to foster innovation, inclusion and lifelong learning through a modern education paradigm. This synthesis highlights the key role of GBL in developing engaging, effective and innovative educational experiences at the dynamic intersection of technology, pedagogy and human-centred design. Educators, policymakers and stakeholders can use GBL principles, practices and capabilities to work together to navigate the evolving educational landscape and create and innovate strategies that empower students and foster creativity for generations.

Finally, while GBL offers transformative potential in education, it has clear limitations that require attention. One of the key issues is the digital divide, where not all students or institutions have equal access to the technology they need, further widening the educational gap. In addition, integrating GBL into established curricula requires careful coordination of learning objectives and assessments, which can lead to potential inconsistencies. Preparing teachers to use GBL effectively is another issue that requires a comprehensive professional development plan. In addition, traditional assessment methods do not adequately account for the nuanced competencies that GBL promotes, requiring innovative assessment strategies. Addressing these challenges requires concerted efforts to promote equal access, improve curriculum integration, strengthen teacher training, update assessment methods and maintain a balanced focus between digital and non-digital play-based activities. Collaborative, interdisciplinary research and partnerships can further develop the theoretical foundations, methodologies and practical applications of GBL to ensure its effectiveness in promoting engaging, inclusive and effective learning experiences for diverse learners.

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REFERENCES

- Ahmadi, A., Noetel, M., Parker, P., Ryan, R. M., Ntoumanis, N., Reeve, J., Beauchamp, M., Dicke, T., Yeung, A., Ahmadi, M., Bartholomew, K., Chiu, T. K. F., Curran, T., Erturan, G., Flunger, B., Frederick, C., Froiland, J. M., González-Cutre, D., Haerens, L., ... & Lonsdale, C. (2023). A Classification System for Teachers' Motivational Behaviors Recommended in Self-Determination Theory Interventions. *Journal of Educational Psychology*, 115(8), 1158–1176. <https://doi.org/10.1037/edu0000783>
- Bandura, A. (2000). Self-Efficacy. *Encyclopedia of Psychology*, 7, 212–213. <https://doi.org/10.1037/10522-094>
- Baker, R. S., & Inventado, P. S. (2018). Educational Data Mining and Learning Analytics. *Foundations of Learning and Instructional Design Technology*, 539–553. <https://doi.org/10.1017/CBO9781139519526.016>
- Becker, K. (2011). Distinctions Between Games and Learning. *Gaming and Simulations*, 75–107. <https://doi.org/10.4018/978-1-60960-195-9.CH105>
- Bell, R. C. (1979). *The Boardgame Book*. Marshall Cavendish.
- Black, P., & Wiliam, D. (2010). Inside the Black Box: Raising Standards Through Classroom Assessment. *Phi Delta Kappan*, 92(1), 81–90. <https://doi.org/10.1177/003172171009200119>
- Caillois, R. (2001). *Man, Play, and Games*. University of Illinois Press.
- Clark, D. B., Tanner-Smith, E. E., & Killingsworth, S. S. (2016). Digital Games, Design, and Learning: A Systematic Review and Meta-Analysis. *Review of Educational Research*, 86(1), 79–122.
- Costa, C., Tyner, K., Henriques, S., & Sousa, C. (2016). A Review of Research Questions, Theories and Methodologies for Game-Based Learning. *Journal of Content, Community & Communication*, 4(2), 2–16.
- Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. Harper & Row.
- de Freitas, S. I. (2006). Using Games and Simulations for Supporting Learning. *Learning, Media and Technology*, 31(4), 343–358. <https://doi.org/10.1080/17439880601021967>
- de la Barrera, U., Mónaco, E., Postigo-Zegarra, S., Gil-Gómez, J. A., & Montoya-Castilla, I. (2021). EmoTIC: Impact of a Game-Based Social-Emotional Programme on Adolescents. *PLoS ONE*, 16(4), e0250384. <https://doi.org/10.1371/journal.pone.0250384>

- Deterding, S., O'Hara, K., Sicart, M., Dixon, D., & Nacke, L. (2011). Gamification: Using Game Design Elements in Non-gaming Contexts. In *Conference on Human Factors in Computing Systems—Proceedings* (pp. 2425–2428). <https://doi.org/10.1145/1979742.1979575>
- Dewey, J. (1977). *John Dewey, the Essential Writings*. Harper & Row.
- Dickey, M. D. (2011). Murder on Grimm Isle: The Impact of Game Narrative Design in an Educational Game-Based Learning Environment. *British Journal of Educational Technology*, 42(3), 456–469. <https://doi.org/10.1111/J.1467-8535.2009.01032.X>
- Dillenbourg, P. (1999). What Do You Mean by Collaborative Learning? In P. Dillenbourg (Ed.), *Collaborative-Learning: Cognitive and Computational Approaches* (pp. 1–19). Elsevier.
- Egenfeldt-Nielsen, S. (2004). Practical Barriers in Using Educational Computer Games. *On the Horizon*, 12(1), 18–21. <https://doi.org/10.1108/10748120410540454>
- Egenfeldt-Nielsen, S. (2006). Overview of Research on the Educational Use of Video Games. *Nordic Journal of Digital Literacy*, 1(3), 184–214. <https://doi.org/10.18261/TSSN1891-943X-2006-03-03>
- Egenfeldt-Nielsen, S., Smith, J. H., & Tosca, S. P. (2020). *Understanding Video Games: The Essential Introduction* (4th ed.). Routledge.
- Fabricatore, C. & López, X. (2014). A Model to Identify Affordances for Game-Based Sustainability Learning. In *Proceedings of the 8th European Conference on Game Based Learning* (pp. 99–109).
- Flavell, J. H. (1979). Metacognition and Cognitive Monitoring: A New Area of Cognitive-Developmental Inquiry. *American Psychologist*, 34(10), 906–911. <https://doi.org/10.1037/0003-066X.34.10.906>
- Freire, M., Serrano-Laguna, Á., Iglesias, B. M., Martínez-Ortiz, I., Moreno-Ger, P., & Fernández-Manjón, B. (2016). Game Learning Analytics: Learning Analytics for Serious Games. *Learning, Design, and Technology*, 1–29. https://doi.org/10.1007/978-3-319-17727-4_21-1
- Gee, J. P. (2003). *What Video Games Have to Teach us About Learning and Literacy*. Palgrave Macmillan.
- Gee, J., & Gee, J. P. (2007). *Social Linguistics and Literacies: Ideology in Discourses* (3rd ed.). Routledge. <https://doi.org/10.4324/9780203944806>
- Göbel, S., Hardy, S., Wendel, V., Mehm, F., & Steinmetz, R. (2010). Serious Games for Health—Personalized Exergames. In *MM'10—Proceedings of the ACM Multimedia 2010 International Conference* (pp. 1663–1666). <https://doi.org/10.1145/1873951.1874316>
- Gobet, F., de Voogt, A., & Retschitzki, J. (2004). *Moves in Mind: The Psychology of Board Games*. Psychology Press.
- Granic, I., Lobel, A., & Engels, R. C. M. E. (2014). The Benefits of Playing Video Games. *The American Psychologist*, 69(1), 66–78. <https://doi.org/10.1037/A0034857>

- Hawlitsek, A., & Joeckel, S. (2017). Increasing the Effectiveness of Digital Educational Games: The Effects of a Learning Instruction on Students' Learning, Motivation and Cognitive Load. *Computers in Human Behavior*, 72, 79–86. <https://doi.org/10.1016/j.chb.2017.01.040>
- Huizinga, J. (1949). *Homo Ludens: A Study of the Play Element in Culture*. Routledge & Kegan Paul (Originally published in 1938).
- Huotari, K., & Hamari, J. (2017). A Definition for Gamification: Anchoring Gamification in the Service Marketing Literature. *Electronic Markets*, 27(1), 21–31. <https://doi.org/10.1007/s12525-015-0212-z>
- Ifenthaler, D., & Yau, J. Y. K. (2020). Utilising Learning Analytics to Support Study Success in Higher Education: A Systematic Review. *Educational Technology Research and Development*, 68(4), 1961–1990. <https://doi.org/10.1007/S11423-020-09788-Z>
- Illeris, K. (2009). *Contemporary Theories of Learning: Learning Theorists ... In Their Own Words*. Routledge.
- Johnson, L., Becker, S. A., Cummins, M., Estrada, V., Freeman, A., & Hall, C. (2016). *NMC Horizon Report: 2016 Higher Education Edition*. The New Media Consortium.
- Kebritchi, M., Hirumi, A., & Bai, H. (2010). The Effects of Modern Mathematics Computer Games on Mathematics Achievement and Class Motivation. *Computers & Education*, 55(2), 427–443. <https://doi.org/10.1016/j.compedu.2010.02.007>
- Klopfer, E., Osterweil, S., & Salen, K. (2009). *Moving Learning Games Forward*. The Education Arcade.
- Kolb, D. A. (1986). *Experiential Learning : Experience as the Source of Learning and Development*. Prentice Hall.
- Koster, R. (2013). *Theory of Fun for Game Design* (2nd ed.). O'Reilly Media.
- Malone, T. W. (1981). Toward a Theory of Intrinsically Motivating Instruction. *Cognitive Science*, 5(4), 333–369. https://doi.org/10.1207/S15516709COG0504_2
- Maratou, V., Ennami, F., Luz, F., Abdullahi, Y., Medeišienė, R. A., Ščiukauskė, I., Chaliampalias, R., Kameas, A. D., Sousa, C., & Rye, S. (2023). Game-Based Learning in Higher Education Using Analogue Games. *International Journal of Film and Media Arts*, 8(1), 68–83. <https://doi.org/10.24140/ijfma.v8.n1.04>
- Mayer, I., Bekebrede, G., Harteveld, C., Warmelink, H., Zhou, Q., van Ruijven, T., Lo, J., Kortmann, R., & Wenzler, I. (2014). The Research and Evaluation of Serious Games: Toward a Comprehensive Methodology. *British Journal of Educational Technology*, 45(3), 502–527. <https://doi.org/10.1111/bjet.12067>

- Mayer, R. E., & Moreno, R. (2003). Nine Ways to Reduce Cognitive Load in Multimedia Learning. *Educational Psychologist*, 38(1), 43–52. https://doi.org/10.1207/S15326985EP3801_6
- Mcdowell, P., Darken, R., Sullivan, J., & Johnson, E. (2006). Delta3D: A Complete Open Source Game and Simulation Engine for Building Military Training Systems. *The Journal of Defense Modeling and Simulation: Applications, Methodology, Technology*, 3(3), 143–154. <https://doi.org/10.1177/154851290600300302>
- Montola, M., Stenros, J., & Waern, A. (2009). *Pervasive Games: Theory and Design*. CRC Press.
- Moon, J. A. (2013). Reflection in Learning and Professional Development: Theory and Practice. *Routledge*. <https://doi.org/10.4324/9780203822296>
- Moreno, R. (Ed.). (2010). Cognitive Load Theory: Historical Development and Relation to Other Theories. In J. L. Plass, R. Moreno, & R. Brünken (Eds.), *Cognitive Load Theory* (pp. 9–28). Cambridge University Press. <https://doi.org/10.1017/CBO9780511844744.003>
- Nadolny, L., & Halabi, A. (2016). Student Participation and Achievement in a Large Lecture Course with Game-Based Learning. *Simulation & Gaming*, 47(1), 51–72. <https://doi.org/10.1177/10468781156203>
- Papert, S. A. (1994). *The Children's Machine: Rethinking School in the Age of the Computer*. Basic Books.
- Parlett, D. (1999). *The Oxford History of Board Games*. Oxford University Press.
- Piaget, J. (1952). *The Origins of Intelligence in Children* (M. Cook, Trans.). W. W. Norton & Co. <https://doi.org/10.1037/11494-000>
- Plass, J. L., Homer, B. D., & Kinzer, C. K. (2015). Foundations of Game-Based Learning. *Educational Psychologist*, 50(4), 258–283. <https://doi.org/10.1080/00461520.2015.1122533>
- Prensky, M. (2003). Digital Game-Based Learning. *Computers in Entertainment (CIE)*, 1(1), 21–21. <https://doi.org/10.1145/950566.95059>
- Przybylski, A. K., Deci, E. L., Rigby, C. S., & Ryan, R. M. (2014). Competence-Impeding Electronic Games and Players' Aggressive Feelings, Thoughts, and Behaviors. *Journal of Personality and Social Psychology*, 106(3), 441–457. <https://doi.org/10.1037/a0034820>
- Rieber, L. P. (2005). The Cambridge Handbook of Multimedia Learning: Multimedia Learning in Games, Simulations, and Microworlds. In *The Cambridge Handbook of Multimedia Learning* (pp. 549–568). <https://doi.org/10.1017/CBO9780511816819.034>
- Rigby, S., & Ryan, R. M. (2011). *Glued to Games: How Video Games Draw Us In and Hold Us Spellbound*. Praeger.
- Riva, G. (2022). Virtual Reality in Clinical Psychology. *Comprehensive Clinical Psychology*, 91. <https://doi.org/10.1016/B978-0-12-818697-8.00006-6>

- Rouse, R. (2005). *Game Design: Theory & Practice* (2nd ed.). Wordware Publishing.
- Ryan, R. M., & Deci, E. L. (2000). Intrinsic and Extrinsic Motivations: Classic Definitions and New Directions. *Contemporary Educational Psychology*, 25(1), 54–67. <https://doi.org/10.1006/CEPS.1999.1020>
- Ryan, R. M., Deci, E. L., Vansteenkiste, M., & Soenens, B. (2021). Building a Science of Motivated Persons: Self-Determination Theory’s Empirical Approach to Human Experience and the Regulation of Behavior. *Motivation Science*, 7(2), 97–110. <https://doi.org/10.1037/mot0000194>
- Ryan, R. M., Rigby, C. S., & Przybylski, A. (2006). The Motivational Pull of Video Games: A Self-determination Theory Approach. *Motivation and Emotion*, 30(4), 347–363. <https://doi.org/10.1007/s11031-006-9051-8>
- Rye, S., & Aktas, E. (2022). Serious Games as a Validation Tool for {PREDIS}: A Decision Support System for Disaster Management. *International Journal of Environmental Research and Public Health*, 19(24), 16584. <https://doi.org/10.3390/ijerph192416584>
- Rye, S., & Sousa, C. (2023). Investigating Inclusivity in Game-Based Learning: Current Practices and Multistakeholder Perspectives. In *Proceedings of the 17th European Conference on Games Based Learning, ECGBL 2023* (pp. 904–911)d. <https://doi.org/10.34190/ecgbl.17.1.1923>
- Salen, K. & Zimmerman, E. (2004). *Rules of Play: Game Design Fundamentals*. MIT Press.
- Schön, D. A. (2017). The Reflective Practitioner: How Professionals Think in Action. *Routledge*. <https://doi.org/10.4324/9781315237473>
- Scouller, K. (1998). The Influence of Assessment Method on Students’ Learning Approaches: Multiple Choice Question Examination Versus Assignment Essay. *Higher Education*, 35, 453–472. <https://doi.org/10.1023/A:1003196224280>
- Shute, V. J., & Rahimi, S. (2017). Review of Computer-Based Assessment for Learning in Elementary and Secondary Education. *Journal of Computer Assisted Learning*, 33(1), 1–19. <https://doi.org/10.1111/jcal.12172>
- Sousa, C., & Costa, C. (2018). Videogames as a Learning Tool: Measuring the efficacy of Game-Based Learning Interventions. *Revista Lusófona de Educação*, 40, 199–210. <https://doi.org/10.24140/issn.1645-7250.rle40.13>
- Sousa, C., Luz, F., Fonseca, M. M., Neves, P., Lopes, P., Maratou, V., Chaliampalias, R., Kameas, A., Abdullahi, Y., & Rye, S. (2022). An Accessible and Inclusive Future for Tabletop Games and Learning: Paradigms and Approaches. In *ICERI2022 Proceedings*. <https://doi.org/10.21125/iceri.2022.2205>
- Sousa, C., Neves, P., & Luz, F. (2023). Barriers and Hindrances to the Effective Use of Games in Education: Systematic Literature Review and Intervention Strategies. *Proceedings of the 17th European Conference on Games Based*

- Learning, ECGBL 2023*, 611–620. <https://doi.org/10.34190/ecgbl.17.1.1472>
- Sousa, C., Rye, S., Sousa, M., Torres, P. J., Perim, C., Mansuklal, S., & Ennami, F. (2023). Playing at the School Table: Systematic Literature Review of Board, Tabletop, and Other Analog Game-Based Learning Approaches. *Frontiers in Psychology*, 14. <https://doi.org/10.3389/fpsyg.2023.1160591>
- Sousa, M., Zagalo, N., & Oliveira, A. P. (2021). Mechanics or Mechanisms: Defining Differences in Analog Games to Support Game Design. In *Proceedings of 2021 IEEE Conference on Games (CoG)* (pp. 1–8). IEEE. <https://doi.org/10.1109/CoG52621.2021.9619055>
- Squire, K. (2006). From Content to Context: Videogames as Designed Experience. *Educational Researcher*, 35(8), 19–29. <https://doi.org/10.3102/0013189X035008019>
- Squire, K. & Jenkins, H. (2011). *Video Games and Learning: Teaching and Participatory Culture in the Digital Age*. Teachers College Press.
- Steinkuehler, C., & Duncan, S. (2008). Scientific Habits of Mind in Virtual Worlds. *Journal of Science Education and Technology*, 17(6), 530–543. <https://doi.org/10.1007/S10956-008-9120-8>
- Sutton-Smith, B. (1997). *The Ambiguity of Play*. Harvard University Press.
- Sweller, J. (1988). Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science*, 12(2), 257–285. https://doi.org/10.1207/s15516709cog1202_4
- Tomlinson, C. A. (2001). How To Differentiate Instruction In *Mixed Ability Classrooms* (2nd ed.). Association for Supervision and Curriculum Development.
- Vygotsky, L. (1978). *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press.
- Wenger, E. (2000). Communities of Practice and Social Learning Systems. *Organization*, 7(2), 225–246. <https://doi.org/10.1177/135050840072002>
- Wertsch, J. V. (1988). *Vygotsky and the Social Formation of Mind*. Harvard University Press.
- Wiley, D. A. (2002). *The Instructional Use of Learning Objects*. Agency for Instructional Technology/Association for Educational Communications & Technology.
- Wirth, J., Stebner, F., Trypke, M., Schuster, C., & Leutner, D. (2020). An Interactive Layers Model of Self-Regulated Learning and Cognitive Load. *Educational Psychology Review*, 32(4), 1127–1149. <https://doi.org/10.1007/s10648-020-09568-4>
- Wouters, P., van Nimwegen, C., van Oostendorp, H., & van Der Spek, E. D. (2013). A Meta-analysis of the Cognitive and Motivational Effects of Serious Games. *Journal of Educational Psychology*, 105(2), 249–265. <https://doi.org/10.1037/a0031311>